Projection onto interval: Real Stability

Research Architecture Runtime
`operator/projection-interval`

2026-07-29

Abstract

`projection-interval` uses interval clamp, which is 1-Lipschitz; hence $|x' - x| \le \varepsilon$ implies the clamped values move by at most ε. Lean `LEAN_FULL`.

This theorem is: `Reduction`.

1 Problem

`projection-interval` projects coordinates onto intervals via $\mathrm{clamp}(x; \mathrm{lo}, \mathrm{hi}) = \max(\mathrm{lo}, \min(x, \mathrm{hi}))$.

2 Stability notion

Nonexpansiveness of clamp: output perturbation $\le$ input perturbation in absolute value.

3 Definitions

Require $\mathrm{lo} \le \mathrm{hi}$. Stability form is the Lipschitz claim with $|x' - x| \le \varepsilon$.

4 Theorem

Theorem 1. *If* $\mathrm{lo} \le \mathrm{hi}$ *and* $|x' - x| \le \varepsilon$*, then* $|\mathrm{clamp}(x') - \mathrm{clamp}(x)| \le \varepsilon$*. The Lipschitz constant* 1 *is sharp.*

5 Intuition

Clipping cannot expand distances: points move weakly toward the interval.

6 Examples

Example 2. $\mathrm{lo} = 0$*,* $\mathrm{hi} = 1$*,* $x = 0.4$*,* $x' = 0.7$*,* $\varepsilon = 0.3$*: both clamp to themselves and move by* 0.3*.*

7 Proof

The map $x \mapsto \max(\mathrm{lo}, \min(x, \mathrm{hi}))$ is 1-Lipschitz on $\mathbb{R}$ when $\mathrm{lo} \le \mathrm{hi}$ (standard case analysis on the three regions $(-\infty, \mathrm{lo}]$, $[\mathrm{lo}, \mathrm{hi}]$, $[\mathrm{hi}, \infty)$). Therefore $|\mathrm{clamp}(x') - \mathrm{clamp}(x)| \le |x' - x| \le \varepsilon$ (Theorem 1). Sharpness: take an interior segment of length ε inside $[\mathrm{lo}, \mathrm{hi}]$.

8 Formal statement

Lean module: `Research.Operators.ProjectionInterval.Preservation`. Source file: `lean/Research/Operator`
Theorems: `projection_interval_clamp_stability`, `projection_interval_clamp_sharpness`. Uses
`Research.Operators.Projection.Clamp`. Certificate: `lean/certificates/projection-interval/projection`
Status: `LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).

9 Proof dependencies

- `Projection.Clamp` nonexpansiveness.
- Case analysis on regions.

10 Consequences

Interval/box/clipping operators reduce to clamp. Related: projection-interval, coordinate-clipping.